

8. Programmable Voltage Detection (PVD)

8.1 Overview

The Programmable Voltage Detection (PVD) module monitors the voltage level input to the VCC pin and the detection level can be selected using a software program. The PVD module consists of several separate voltage level detectors which measure the voltage level input to the VCC pin. PVD registers allow your application to configure detection of VCC change at various voltage thresholds.

Each voltage level detector has a voltage monitor associated with it. Voltage monitor registers are used to configure the PVD to trigger an interrupt, event link output, or reset when the thresholds are crossed.

Table 8.1 lists the PVD specifications. Figure 8.1 shows a block diagram of the voltage monitor 0 reset generation circuit. Figure 8.2 and Figure 8.3 show a block diagram of the voltage monitor m/n interrupt and reset circuit ($m = 1, 2, n = 4, 5$).

The PVD module consists of low voltage detection (LVD) and High voltage detection (HVD). LVD detects that VCC is dropping below the voltage threshold. HVD detects that VCC is rising above the voltage threshold.

PVDn are implemented for security purposes. Compared to PVDm, the functions available are more limited. PVDn has a register lock function for security.

The voltage level detector for core voltage is implemented. It monitors the voltage level input to the VDD, and the detection level is pre-defined by hardware. The voltage level detector for core voltage has a voltage monitor circuit associated with it. The voltage monitor circuit for core voltage can assert the reset (Core Voltage Monitor Reset) when the core voltage crossed the threshold as under Vdet_VDDL and over Vdet_VDDH.

The function of Core Voltage Monitor Reset (CVMR) is enabled when OFS2.CVMRDIS = 0.

Table 8.1 PVD specifications (1 of 2)

Item		Programable voltage monitor 0	Programable voltage monitor m ($m = 1, 2$)	Programable voltage monitor n ($n = 4, 5$)
Voltage Monitoring	Monitored pin	VCC	VCC	VCC
	Detected Event	Voltage falls past V_{det0}	Voltage rises or falls past V_{detm}	Voltage rises or falls past V_{detn}
	Detection Voltage	Selectable from among 8 different levels by using OFS1.VDSEL0 [2:0] bits as V_{det0}	Selectable from among 16 different levels by using PVDmCMPCR.PVDLVL[4:0] bits as V_{detm}	Selectable from among 16 different levels by using PVDnCMPCR.PVDLVL[4:0] bits as V_{detn}
	Monitor Flag	None	PVDmSR.MON flag: Monitors whether voltage is higher or lower than V_{detm} PVDmSR.DET flag: V_{detm} passage detection	None
Process on voltage detection	Reset	Voltage monitor 0 reset	Voltage monitor m reset	Voltage monitor n reset
		Reset when $V_{det0} > VCC$ CPU restart after specified time with VCC $> V_{det0}$	Reset condition selectable: Reset when $V_{detm} > VCC$ (LVD) or $V_{detm} < VCC$ (HVD) CPU restart timing selectable: after specified time with LVD or HVD	Reset condition selectable: Reset when $V_{detn} > VCC$ (LVD) or $V_{detn} < VCC$ (HVD)
	Interrupt	No interrupt	Voltage monitor m interrupt Non-maskable interrupt or maskable interrupt selectable Interrupt request issued in response to either or both LVD and HVD	No interrupt
Digital filter	Enable/Disable Switching	Digital filter function not available	Available	Available
	Sampling Time	—	1/s LOCO frequency $\times 2$ (s: 2, 4, 8, 16)	1/s LOCO frequency $\times 2$ (s: 2, 4, 8, 16)
Event linking		None	Available Output of event signals on detection of V_{detm} crossings	None

Table 8.1 PVD specifications (2 of 2)

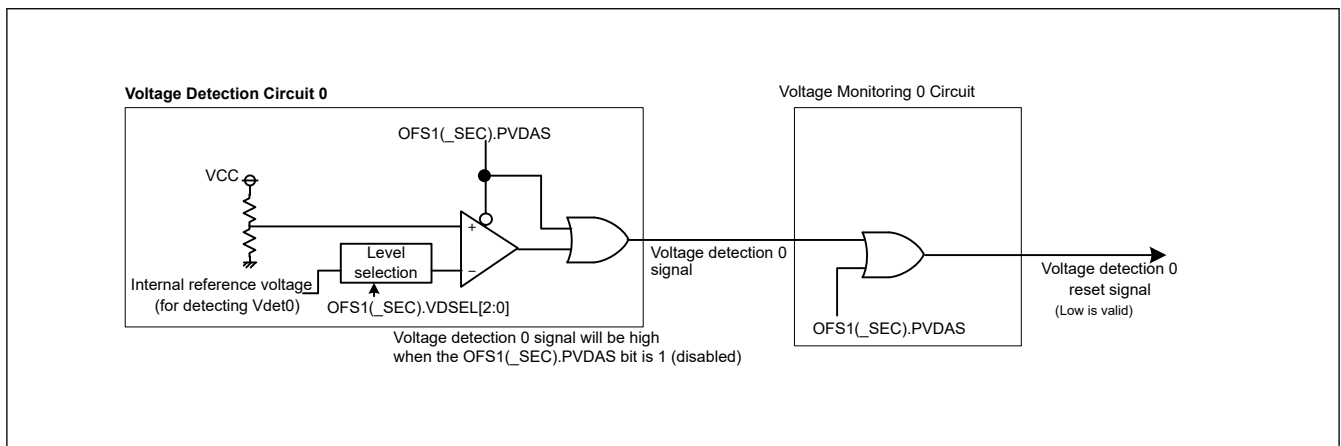
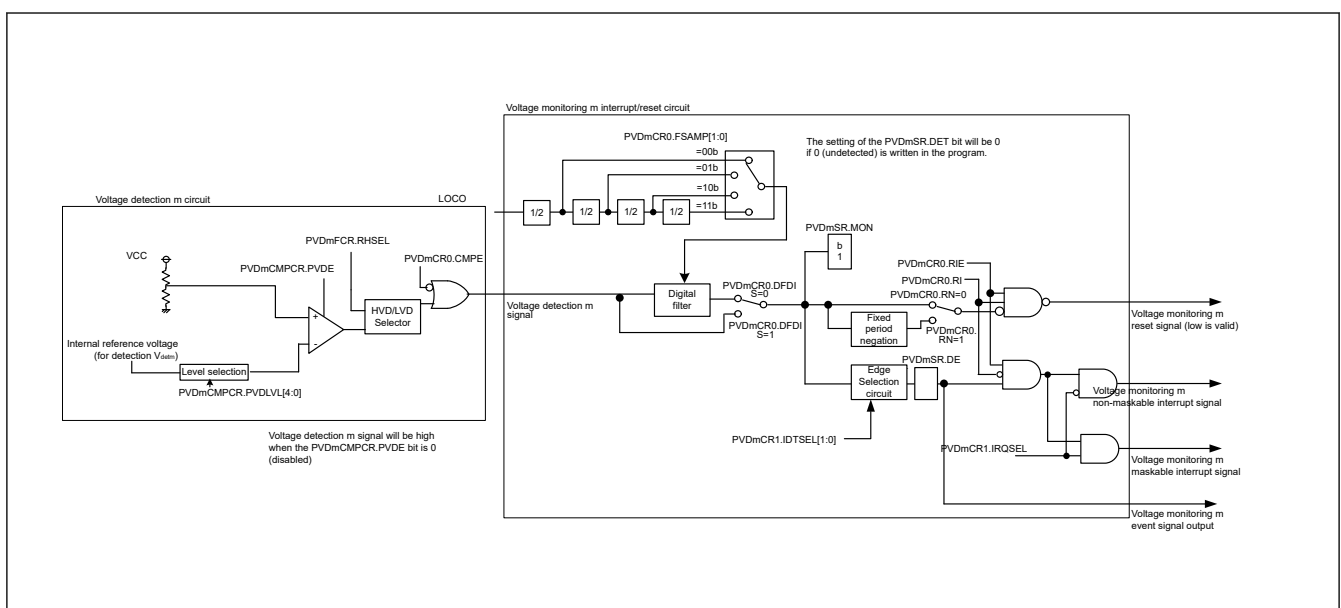
Item	Programable voltage monitor 0	Programable voltage monitor m (m = 1, 2)	Programable voltage monitor n (n = 4, 5)
Register lock	None	None	PVDLR

Table 8.2 CVM specifications

Item	Core voltage monitor	
Voltage Monitoring	Monitored pin	VDD
	Detected Event	Voltage falls past V_{det_VDDL} Voltage rises past V_{det_VDDH}
	Detection Voltage	Pre-defined level*1
Process on voltage detection	Reset	Core Voltage monitor reset*2
		Reset when $V_{det_VDDL} > VDD$ or $V_{det_VDDH} < VDD$ CPU restart after specified time with $V_{det_VDDL} \leq VDD \leq V_{det_VDDH}$

Note 1. It means V_{det_VDDL} and V_{det_VDDH}

Note 2. See the chapter of Reset about core voltage monitor reset.

**Figure 8.1 Block diagram of voltage monitor 0 reset circuit****Figure 8.2 Block diagram of voltage monitor m interrupt/reset circuit (m = 1, 2. The m is integer).**

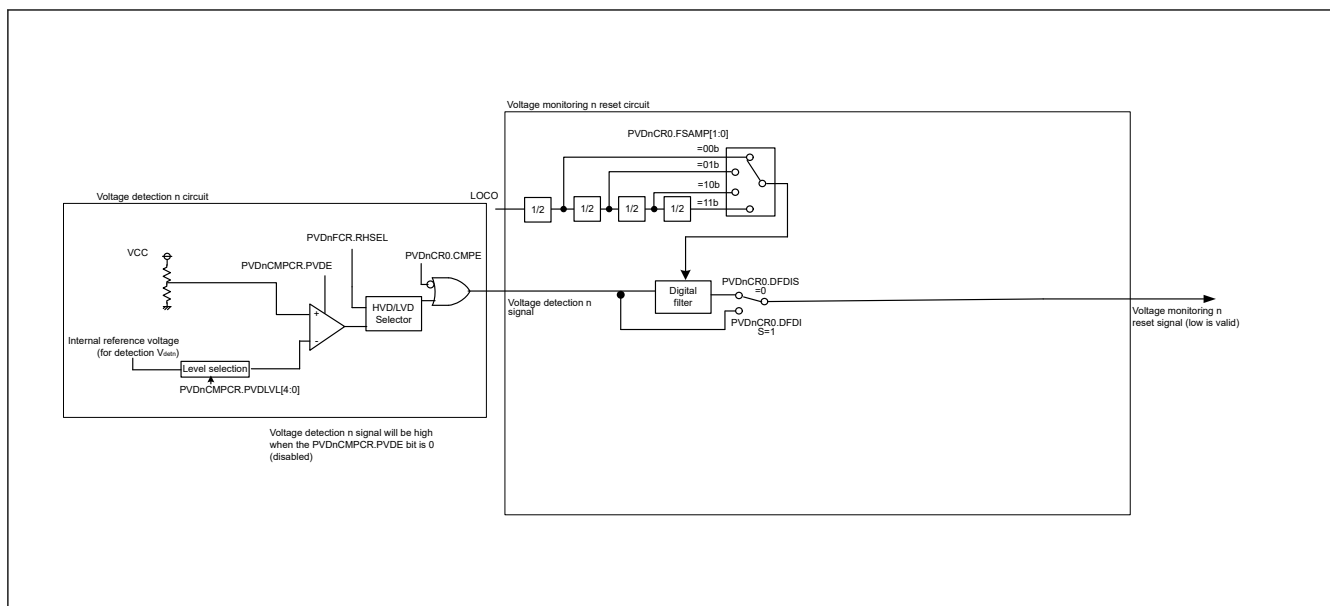


Figure 8.3 Block diagram of voltage monitor n reset circuit (n = 4, 5. The n is integer).

8.2 Register Descriptions

8.2.1 PVDSAR : Programable Voltage Detection Security Attribution Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3CC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	NONS EC1	NONS EC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NONSEC0	Non Secure Attribute bit 0 Target register: registers for PVD1 0: Secure 1: Non Secure	R/W
1	NONSEC1	Non Secure Attribute bit 1 Target register: registers for PVD2 0: Secure 1: Non Secure	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

Note: This register is write-protected by PRCR register.

The PVDSAR register controls the secure attribute of PVD registers.

NONSEC0 bit (Non Secure Attribute bit 0)

This bit controls the security attribute of PVD1CMPCR, PVD1CR0, PVD1CR1, PVD1SR.

NONSEC1 bit (Non Secure Attribute bit 1)

This bit controls the security attribute of PVD2CMPCR, PVD2CR0, PVD2CR1, PVD2SR.

8.2.2 PVDmCMPCR : Voltage Monitor m Comparator Control Register (m = 1, 2)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA58 + 0x4 × (m - 1)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	PVDE	—	—	PVDLVL[4:0]				
Value after reset:	0	0	0	0	1	1	1	1

Bit	Symbol	Function	R/W
4:0	PVDLVL[4:0]	Detection Voltage m Level Select Settings other than the following are prohibited. These are standard voltage level during the fall of voltage. For details on standard voltage level during the rise of voltage, see section 60, Electrical Characteristics . 0x03: 3.86 V (Vdetm_3) 0x04: 3.14 V (Vdetm_4) 0x05: 3.10 V (Vdetm_5) 0x06: 3.08 V (Vdetm_6) 0x07: 2.85 V (Vdetm_7) 0x08: 2.83 V (Vdetm_8) 0x09: 2.80 V (Vdetm_9) 0x0A: 2.62V (Vdetm_10) 0x0B: 2.33V (Vdetm_11) 0x0C: 1.90V (Vdetm_12) 0x0D: 1.86V (Vdetm_13) 0x0E: 1.74V (Vdetm_14) 0x0F: 1.71V (Vdetm_15) Others: Setting prohibited	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	PVDE	Voltage Detection m Enable 0: Voltage detection m circuit disabled 1: Voltage detection m circuit enabled	R/W

Note: S-TYPE-3, P-TYPE-2

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

PVDLVL[4:0] bits (Detection Voltage m Level Select)

The PVDmCMPCR.PVDLVL can be changed only if all the PVDmCMPCR.PVDE bits and all the PVDnCMPCR.PVDE bits are 0. All the voltage detection circuits should not be set at the same voltage detection level.

Do not change PVDmCMPCR.PVDLVL and PVDmCMPCR.PVDE at the same time.

PVDE bit (Voltage Detection m Enable)

When using voltage detection m interrupt/reset or the PVDmSR.MON bit, set the PVDmCMPCR.PVDE bit to 1. The voltage detection m circuit starts once td(E-A) passes after the PVDmCMPCR.PVDE bit value is changed from 0 to 1. When using the voltage detection m circuit in Deep Software Standby mode, do not transit to Deep Software Standby mode 3.

8.2.3 PVDnCMPCR : Voltage Monitor n Comparator Control Register (n = 4, 5)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA64 + 0x4 × (n - 4)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	PVDE	—	—	PVDLVL[4:0]				

Value after reset: 0 0 0 0 1 1 1 1

Bit	Symbol	Function	R/W
4:0	PVDLVL[4:0]	Detection Voltage n Level Select Settings other than above are prohibited. (These are standard voltage level during fall of voltage. About standard voltage level during rise of voltage, refer to chapter of electrical characteristics.) 0 0 0 1 1 : 3.86V (Vdetn_3) 0 0 1 0 0 : 3.14V (Vdetn_4) 0 0 1 0 1 : 3.10V (Vdetn_5) 0 0 1 1 0 : 3.08V (Vdetn_6) 0 0 1 1 1 : 2.85V (Vdetn_7) 0 1 0 0 0 : 2.83V (Vdetn_8) 0 1 0 0 1 : 2.80V (Vdetn_9) 0 1 0 1 0 : 2.62V (Vdetn_10) 0 1 0 1 1 : 2.33V (Vdetn_11) 0 1 1 0 0 : 1.90V (Vdetn_12) 0 1 1 0 1 : 1.86V (Vdetn_13) 0 1 1 1 0 : 1.74V (Vdetn_14) 0 1 1 1 1 : 1.71V (Vdetn_15)	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	PVDE	Voltage Detection n Enable 0: Voltage detection n circuit disabled 1: Voltage detection n circuit enabled	R/W

Note: S-TYPE-6, P-TYPE-2

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

The PVDnCMPCR.PVDLVL can be changed only if all the PVDmCMPCR.PVDE bits and all the PVDnCMPCR.PVDE bits are 0. All the voltage detection circuits should not be set at the same voltage detection level.

Please do not change PVDnCMPCR.PVDLVL and PVDnCMPCR.PVDE at the same time.

PVDE bit (Voltage Detection n Enable)

When using voltage detection n interrupt/reset or the PVDnSR.MON bit, set the PVDnCMPCR.PVDE bit to 1. The voltage detection n circuit starts once td(E-A) passes after the PVDnCMPCR.PVDE bit value is changed from 0 to 1. When using the voltage detection n circuit in deep software standby mode, do not transit to deep software standby mode 3.

8.2.4 PVDmCR0 : Voltage Monitor m Circuit Control Register 0 (m = 1, 2)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA70 + 0x4 × (m - 1)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	RN	RI	FSAMP[1:0]	—	CMPE	DFDIS	RIE	

Value after reset: 1 0 0 0 x 0 1 0

Bit	Symbol	Function	R/W
0	RIE	Voltage Monitor m Interrupt/Reset Enable 0: Disable 1: Enable	R/W
1	DFDIS	Voltage monitor m Digital Filter Disabled Mode Select 0: Enable the digital filter 1: Disable the digital filter	R/W
2	CMPE	Voltage Monitor m Circuit Comparison Result Output Enable 0: Voltage monitor m circuit comparison result output disabled 1: Voltage monitor m circuit comparison result output enabled	R/W
3	—	The read value is undefined. The write value should be 1.	R/W
5:4	FSAMP[1:0]	Sampling Clock Select 0 0: 1/2 LOCO frequency 0 1: 1/4 LOCO frequency 1 0: 1/8 LOCO frequency 1 1: 1/16 LOCO frequency	R/W
6	RI	Voltage Monitor m Circuit Mode Select (1) case of PVDmFCR.RHSEL = 0 0: Voltage monitor m interrupt during V_{detm} passage 1: Voltage monitor m reset enabled when the voltage falls to and below V_{detm} (2) case of PVDmFCR.RHSEL = 1 0: Prohibited 1: Voltage monitoring m reset enabled when the voltage rises to and up V_{detm}	R/W
7	RN	Voltage Monitor m Reset Negate Select (1) case of PVDmFCR.RHSEL = 0 0: Negation follows a stabilization time (t_{pVDM}) after $VCC > V_{detm}$ is detected. 1: Negation follows a stabilization time (t_{pVDM}) after assertion of the PVDm reset. (2) case of PVDmFCR.RHSEL = 1 0: Negation follows a stabilization time (t_{pVDM}) after $VCC < V_{detm}$ is detected. 1: Prohibited	R/W

Note: S-TYPE-3, P-TYPE-2

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

RIE bit (Voltage Monitor m Interrupt/Reset Enable)

The RIE bit enables or disables voltage monitor m interrupt/reset. Ensure that neither a voltage monitor m interrupt nor a voltage monitor m reset is generated during programming or erasure of the MRAM.

DFDIS bit (Voltage monitor m Digital Filter Disabled Mode Select)

The DFDIS bit disables the digital filter circuit. Set the LOCOCR.LCSTP bit to 0 (the LOCO operates) if the DFDIS bit is 0 (enabling the digital filter circuit). Set the DFDIS bit to 1 (digital filter circuit disabled) when using voltage monitor m circuit in software standby mode or Deep Software Standby mode.

CMPE bit (Voltage Monitor m Circuit Comparison Result Output Enable)

The CMPE bit enables or disables voltage monitor m circuit comparison result output. Set the CMPE bit to 1 after the voltage detection m circuit enables and stabilization time ($t_{d(E-A)}$) elapses. When stopping the voltage detection m circuit, disable the voltage detection m circuit after setting the CMPE bit is 0.

FSAMP[1:0] bits (Sampling Clock Select)

The FSAMP[1:0] bits can be rewritten only when the PVDmCR0.DFDIS bit is 1 (digital filter circuit disabled). The FSAMP[1:0] bits should not be modified when the PVDmCR0.DFDIS bit is 0 (digital filter circuit enabled).

RI bit (Voltage Monitor m Circuit Mode Select)

When the PVDmCR0.RI bit is 1 (voltage monitor m reset selected), a transition to Deep Software Standby mode 2 or 3 cannot be made, instead a transition to Deep Software Standby mode 1 is made. To enter Deep Software Standby mode 2 or 3, set all the PVDmCR0.RI bits to 0 (voltage monitor m interrupt selected).

RN bit (Voltage Monitor n Reset Negate Select)

Case of PVDmFCR.RHSEL = 0

If the RN bit is set to 1 (negation follows a stabilization time after assertion of the PVDm reset signal), set the LOCOCR.LCSTP bit to 0 (the LOCO operates). Furthermore, if a transition to Software Standby mode or Deep Software Standby mode is to be made, the only possible value for the RN bit is 0 (negation follows a stabilization time after $V_{CC} > V_{detm}$ is detected). Do not set the RN bit to 1 (negation follows a stabilization time after assertion of the PVDm reset signal) when this is the case.

Case of PVDmFCR.RHSEL = 1

Do not set the RN bit to 1.

8.2.5 PVDnCR0 : Voltage Monitor n Circuit Control Register 0 (n = 4, 5)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA7C + 0x4 × (n - 4)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	FSAMP[1:0]	—	CMPE	DFDIS	RE	
Value after reset:	0	1	0	0	x	0	1	0

Bit	Symbol	Function	R/W
0	RE	Voltage Monitor n Reset Enable 0: Disabled 1: Enabled	R/W
1	DFDIS	Voltage Monitor n Digital Filter Disable Mode Select 0: Digital filter enabled 1: Digital filter disabled	R/W
2	CMPE	Voltage Monitor n Circuit Comparison Result Output Enable 0: Voltage monitor n circuit comparison result output disabled. 1: Voltage monitor n circuit comparison result output enabled.	R/W
3	—	The read value is undefined. The write value should be 1.	R/W
5:4	FSAMP[1:0]	Sampling Clock Select 0 0: 1/2 LOCO frequency 0 1: 1/4 LOCO frequency 1 0: 1/8 LOCO frequency 1 1: 1/16 LOCO frequency	R/W
6	—	This bit is read as 1. The write value should be 1.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

RE bit (Voltage Monitor n Reset Enable)

RE bit decides whether to enable Voltage Monitor n Reset. When RE bit is 1, this function is activated. Ensure that a voltage Monitor n reset isn't generated during programming or erasure of the MRAM.

DFDIS bit (Voltage Monitor n Digital Filter Disable Mode Select)

Set the LOCOCR.LCSTP bit to 0 (the LOCO operates) if the DFDIS bit is 0 (enabling the digital filter circuit).

Set the DFDIS bit to 1 (digital filter circuit disabled) when using voltage monitor n circuit in software standby mode or deep software standby mode.

FSAMP[1:0] bits (Sampling Clock Select)

The FSAMP[1:0] bits can be modified only when the PVDnCR0.DFDIS bit is 1 (digital filter circuit disabled). The FSAMP[1:0] bits should not be modified when the PVDnCR0.DFDIS bit is 0 (digital filter circuit enabled).

8.2.6 PVDmCR1 : Voltage Monitor m Circuit Control Register (m = 1, 2)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0E0 + 0x2 × (m-1)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	IRQSE L	IDTSEL[1:0]	
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
1:0	IDTSEL[1:0]	Voltage Monitor m Interrupt Generation Condition Select (1) case of PVDmFCR.RHSEL = 0 0 0: When $VCC \geq V_{detm}$ (rise) is detected 0 1: When $VCC < V_{detm}$ (fall) is detected 1 0: When fall and rise are detected 1 1: Settings prohibited (2) case of PVDmFCR.RHSEL = 1 This bit is not applicable.	R/W
2	IRQSEL	Voltage Monitor m Interrupt Type Select 0: Non-maskable interrupt 1: Maskable interrupt*1	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note: Set the PRCR.PRC3 bit to 1 (writing enabled) before rewriting this register.

Note 1. When enabling maskable interrupts, do not change the value of the NMIER.PVDmEN bit on the ICU side from the reset state.

8.2.7 PVDmSR : Voltage Monitor m Circuit Status Register (m = 1, 2)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x0E1 + 0x2 × (m - 1)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MON	DET
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
0	DET	Voltage Monitor m Voltage Change Detection Flag*1 (1) case of PVDmFCR.RHSEL = 0 0: Not detected 1: V_{detm} passage detection (2) case of PVDmFCR.RHSEL = 1 This bit is not applicable.*2	R/W
1	MON	Voltage Monitor 1 Signal Monitor Flag (1) case of PVDmFCR.RHSEL = 0 0: $VCC \leq V_{detm}$ 1: $VCC > V_{detm}$ or MON is disabled (2) case of PVDmFCR.RHSEL = 1 0: $VCC \geq V_{detm}$ 1: $VCC < V_{detm}$ or MON is disabled	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note: Set the PRCR.PRC3 bit to 1 (write enabled) before rewriting this register.

Note 1. Only 0 can be written to this bit. After writing 0 to this bit, 2 system clock cycles are required for the bit to be read as 0.

Note 2. DET is depended on PVDmCR1.IDTSEL[1:0].

PVDmCR1.IDTSEL[1:0] = 0 0: When $VCC < V_{detm}$ (fall) is detected, DET is set.

PVDmCR1.IDETSEL[1:0] = 0 1: When VCC ≥ Vdetm (rise) is detected, DET is set.

PVDmCR1.IDETSEL[1:0] = 1 0: When fall or rise are detected, DET is set.

PVDmCR1.IDETSEL[1:0] = 1 1: Settings prohibited

DET flag (Voltage Monitor m Voltage Change Detection Flag)

The DET flag is enabled when the PVDmCMPCR.PVDE bit is 1 (voltage detection m circuit enabled) and the PVDmCR0.CMPE bit is 1 (voltage monitor m circuit comparison result output enabled).

The DET flag should be set to 0 after PVDmCR0.RIE is set to 0 (disabled). PVDmCR0.RIE can be set to 1 (enabled) after a period of two or more cycles of PCLKB has elapsed.

Depending on the number of cycles of PCLKB defined for access to read an I/O register, two or more cycles than PCLKB may have to be secured as waiting time.

MON flag (Voltage Monitor 1 Signal Monitor Flag)

The MON flag is enabled when the PVDmCMPCR.PVDE bit is 1 (voltage detection m circuit enabled) and the PVDmCR0.CMPE bit is 1 (voltage monitor m circuit comparison result output enabled).

8.2.8 PVDmFCR : Voltage Monitor m Function Control Register (m = 1, 2)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB20 + 0x4 × (m - 1)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	RHSEL

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	RHSEL	Rise Hysteresis Select 0: Hysteresis level for VCC-fall detection is selected. 1: Hysteresis level for VCC-rise detection is selected.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

RHSEL bit (Rise Hysteresis Select)

When RHSEL = 0 for VCC-fall detection, the hysteresis level is set above the voltage detection level set by PVDLVL[4:0].
When RHSEL = 1 for VCC-rise detection, the hysteresis level is set under the voltage detection level set by PVDLVL[4:0].
See the chapter of electrical characteristics about the detail of hysteresis level of PVD.

When PVD reset generated by the VCC-fall detection is required, you should set RHSEL to 0.

When PVD reset generated by the VCC-rise detection is required, you should clear RHSEL to 1.

RHSEL must not be set to 1 when PVDmCR0.RI = 0.

RHSEL can be changed only if all the PVDmCMPCR.PVDE bits and all the PVDnCMPCR.PVDE bits are 0.

8.2.9 PVDnFCR : Voltage Monitor n Function Control Register (n = 4, 5)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB2C + 0x4 × (n - 1)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	RHSEL

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	RHSEL	Rise Hysteresis Select 0: Hysteresis level for VCC-fall detection is selected. 1: Hysteresis level for VCC-rise detection is selected.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

RHSEL bit (Rise Hysteresis Select)

When RHSEL = 0 for VCC-fall detection, the hysteresis level is set above the voltage detection level set by PVDLVL[4:0].
When RHSEL = 1 for VCC-rise detection, the hysteresis level is set under the voltage detection level set by PVDLVL[4:0].
See the chapter of electrical characteristics about the detail of hysteresis level of PVD.

When PVD reset generated by the VCC-fall detection is required, you should set RHSEL to 0.

When PVD reset generated by the VCC-rise detection is required, you should clear RHSEL to 1.

RHSEL can be changed only if all the PVDmCMPCR.PVDE bits and all the PVDnCMPCR.PVDE bits are 0.

8.2.10 PVDLR : Voltage Monitor Lock Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xB34

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LOCK

Value after reset: 0 0 0 0 0 0 0 0 1

Bit	Symbol	Function	R/W
0	LOCK	LOCK control 0: The control registers for PVD4 and PVD5 can be written. 1: The control registers for PVD4 and PVD5 can not be written.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6, P-TYPE-2

This register is initialized by POR, RES pin reset, Voltage Monitor 0 reset (PVD0 reset).

LOCK bit (LOCK control)

The initial value of LOCK bit is 1. When the LOCK bit is 1, the control registers for PVD4 and PVD5 can not be written. If the LOCK bit is written to 0 after the LOCK is initialized by the resets which can initialize the LOCK, the control registers for PVD4 and PVD5 can be written.

After that, if you write an arbitrary value to the LOCK, the LOCK bit is fixed to 1. Then the control registers for PVD4 and PVD5 can not be written.

Here, the control registers for PVD4 and PVD5 are PVD4CMPCR, PVD5CMPCR, PVD4CR0, PVD5CR0, PVD4FCR, PVD5FCR.

8.3 VCC Input Voltage Monitor

8.3.1 Monitoring V_{det0}

The comparison results from voltage monitor 0 are not available for reading.

8.3.2 Monitoring V_{detm}

After the settings are completed, results of comparison by voltage Monitor m can be monitored by using the PVDmSR.MON flag.

Table 8.3 Procedures to set up monitoring against V_{detm}

Step		Monitoring the comparison results from voltage monitor 1
Setting the voltage detection m circuit	1	Set PVDmCMPCR.PVDE = 0 to disable voltage detection m before writing to the PVDmCMPCR.PVDLVL[4:0] bits.
	2	Select the detection voltage in the PVDmCMPCR.PVDLVL[4:0] bits.
	3	Set PVDmCMPCR.PVDE = 1 to enable the voltage detection m circuit.
	4	Wait for at least $t_{d(E-A)}$ *1.
Setting the digital filter*2	5	Select the sampling clock for the digital filter in the PVDmCR0.FSAMP[1:0] bits.
	6	Set PVDmCR0.DFDIS = 0 to enable the digital filter.
	7	Wait for at least $2s + 3$ cycles of the LOCO (where $s = 2, 4, 8, 16$, and the sampling clock for the digital filter is the LOCO frequency-divided by s).
Enabling output	8	Set PVDmCR0.CMPE = 1 to enable output of the comparison results from voltage monitor m.

Note 1. Steps 5 to 7 can be performed during the waiting time of step 4.

Note 2. Steps 5 to 7 are not required if the digital filter is not in use.

8.4 Reset from Voltage Monitor 0

When using the reset from voltage monitor 0, clear the OFS1(_SEC).PVDAS bit to 0. (enabling the voltage monitor 0 reset after a reset)

Figure 8.4 shows an example of operations for a voltage monitor 0 reset.

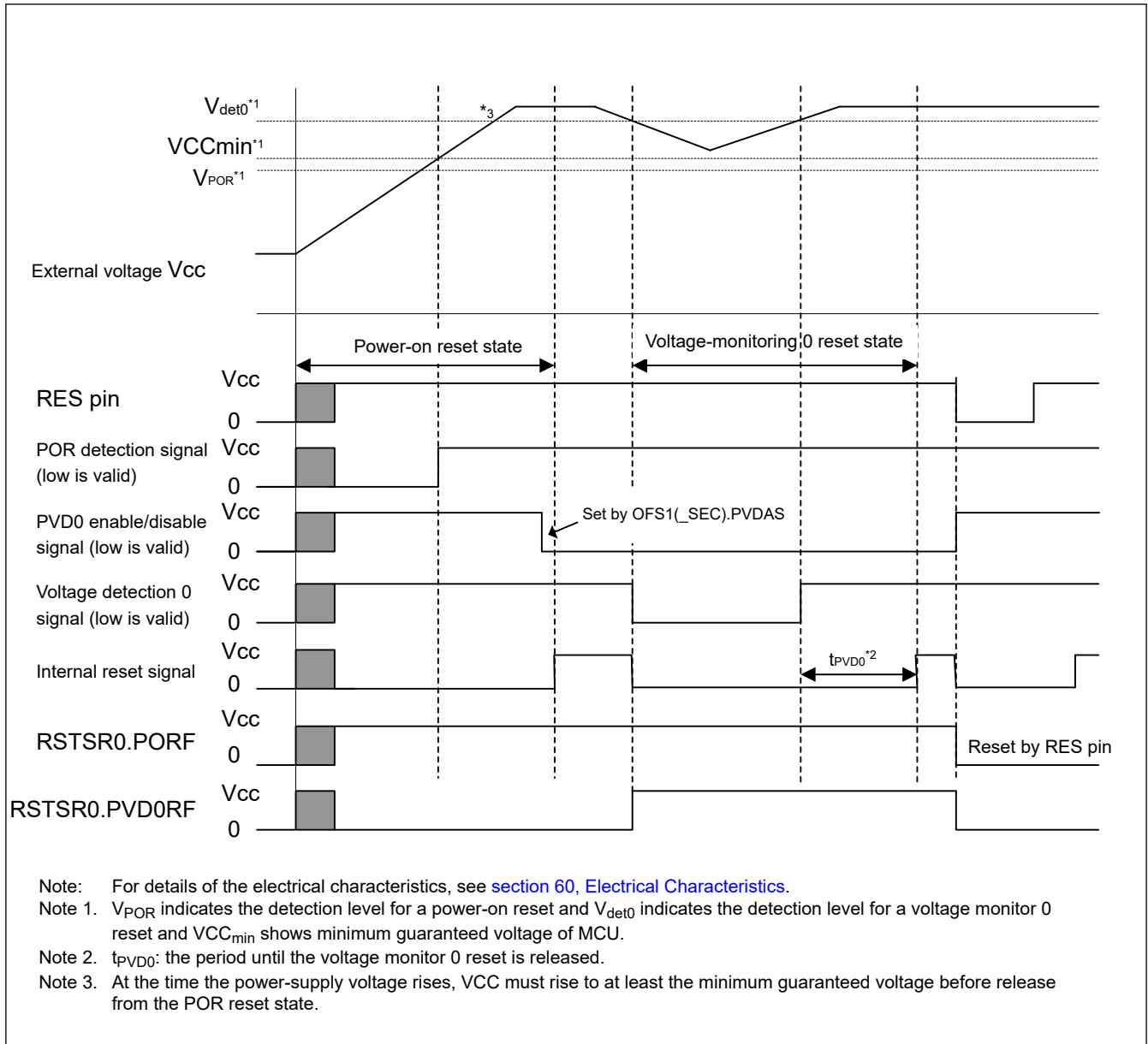


Figure 8.4 Example of voltage monitor 0 reset operation

8.5 Interrupt and Reset from Voltage Monitor m (m = 1, 2)

An interrupt or reset can be generated in response to the results of comparison by the voltage detection m circuit

[Table 8.4](#) shows the procedures for setting bits related to the voltage monitor m interrupt and voltage monitor m reset so that voltage monitoring operates. [Table 8.5](#) shows the procedures for setting bits related to the voltage monitor m interrupt and voltage monitor m reset so that voltage monitoring stops. [Figure 8.5](#) shows an example of operations for a voltage monitor m interrupt. For the operation of the voltage monitor m reset, see [Figure 6.2](#) in [section 6, Resets](#).

Furthermore, if you intend to use the voltage Monitor m circuit in software standby or Deep Software Standby mode, make settings for the voltage Monitor m circuit according to the following procedures.

(1) Setting in Software Standby mode

- Disable the digital filter ($PVDmCR0.DFDIS = 1$).
- After $VCC > V_{detm}$ is detected, negate the voltage monitor m reset signal ($PVDmCR0.RN = 0$) following a stabilization time.

(2) Settings in Deep Software Standby mode

- Disable the digital filter (PVDmCR0.DFDIS = 1).
- Enable the voltage monitor m interrupt (PVDmCR0.RI = 0). If the voltage monitor m reset is enabled (PVDmCR0.RI = 1), a transition to Deep Software Standby mode 2 or 3 is not possible, and the transition will be to Deep Software Standby mode 1 instead.
- During Deep Software Standby mode 3, the voltage Monitor m circuit is stopped. If you intend to use the voltage Monitor m circuit in Deep Software Standby mode, do not transit to Deep Software Standby mode 3.

(3) Disabling at on-chip debug mode

- The function of the voltage Monitor m circuit is disabled by setting DBGSTOPCR.DBGSTOP_PVD at on-chip debug mode.

Table 8.4 Procedures for setting bits related to voltage monitor m interrupt and voltage monitor m reset so that voltage monitoring operates (m = 1, 2)

Step	Voltage monitor m interrupt (voltage monitor m ELC event output)	Voltage monitor m reset
Setting up the voltage detection m circuit	1 Set PVDmCMPCR.PVDE = 0 (disable voltage detection m before writing to the PVDmCMPCR register.)	
	2 Select the detection voltage by setting the PVDmCMPCR.PVDLVL bits.	
	3 —	Select the type of the reset condition by setting the PVDmFCR.RHSEL bit.
	4 Set PVDmCMPCR.PVDE = 1 (enabling the voltage detection m circuit).	
	5 Wait for at least $t_d(E-A)^{*1}$.	
Setting the digital filter ^{*2}	6 Select the sampling clock for the digital filter by the setting PVDmCR0.FSAMP[1:0] bits.	
	7 Set PVDmCR0.DFDIS = 0 (enabling the digital filter).	
	8 Wait for at least $2s + 3$ cycles of the LOCO (where s = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by s) ^{*4} .	
Setting up the voltage monitor m interrupt or reset	9 Set PVDmCR0.RI = 0 to select the voltage monitor m interrupt.	• Set PVDmCR0.RI = 1 (selecting the voltage monitor m reset). • Select the type of the reset negation by setting the PVDmCR0.RN bit.
	10 • Select the timing of interrupt requests by setting the PVDmCR1.IDTSEL[1:0] bits. • Select the type of interrupt by setting the PVDmCR1.IRQSEL bit.	—
Enabling output	11 Set PVDmSR.DET = 0.	
	12 Set PVDmCR0.RIE = 1 (enabling the voltage Monitor m interrupt or reset) ^{*3} .	
	13 Set PVDmCR0.CMPE = 1 (enabling output of the results of comparison by voltage monitor m).	

Note 1. Steps 6 to 12 can be performed during the wait time in step 5.

Note 2. Steps 6 to 8 are not required if the digital filter is not in use.

Note 3. Step 12 is not required if only the ELC event signal is to be output.

Note 4. Steps 9 to 12 can be performed during the waiting time of step 8.

Table 8.5 Procedures for setting bits related to voltage monitor m interrupt and voltage monitor m reset so that voltage monitoring stops (m = 1, 2) (1 of 2)

Step	Voltage monitor m interrupt (voltage monitor m ELC event output), voltage monitor m reset
Stopping the enabling output	1 Set PVDmCR0.CMPE = 0 (disabling output of the results of comparison by voltage Monitor m).
	2 Wait for at least $2s + 3$ cycles of the LOCO (where s = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by s) ^{*1} .
	3 Set PVDmCR0.RIE = 0 (disabling the voltage monitor m interrupt or reset) ^{*3} .
Stopping the digital filter	4 Set PVDmCMPCR.DFDIS = 1 (disabling the digital filter) ^{*1 *2} .

Table 8.5 Procedures for setting bits related to voltage monitor m interrupt and voltage monitor m reset so that voltage monitoring stops (m = 1, 2) (2 of 2)

Step		Voltage monitor m interrupt (voltage monitor m ELC event output), voltage monitor m reset
Stopping the voltage detection m circuit	5	Set PVDmCMPCR.PVDE = 0 (disabling the voltage detection m circuit).

Note 1. Steps 2 and 4 are not required if the digital filter is not in use.

Note 2. To disable the digital filter from its enabled state and then re-enable it, disable it and wait for at least two cycles of the LOCO before re-enabling it.

Note 3. Step 3 is not required if only the ELC event signal is to be output.

If the voltage monitor m interrupt or voltage monitor m reset setting is to be made again after it has been used and stopped once, the following steps in the procedures for stopping and making the setting can be omitted according to the condition:

- Setting or stopping the voltage detection m circuit is not required if the setting for the voltage detection m circuit is not to be changed
- Setting or stopping the digital filter is not required if the setting for the digital filter is not to be changed
- Setting the voltage monitor m interrupt or reset is not required if the setting for the voltage monitor m interrupt or voltage monitor m reset is not to be changed.

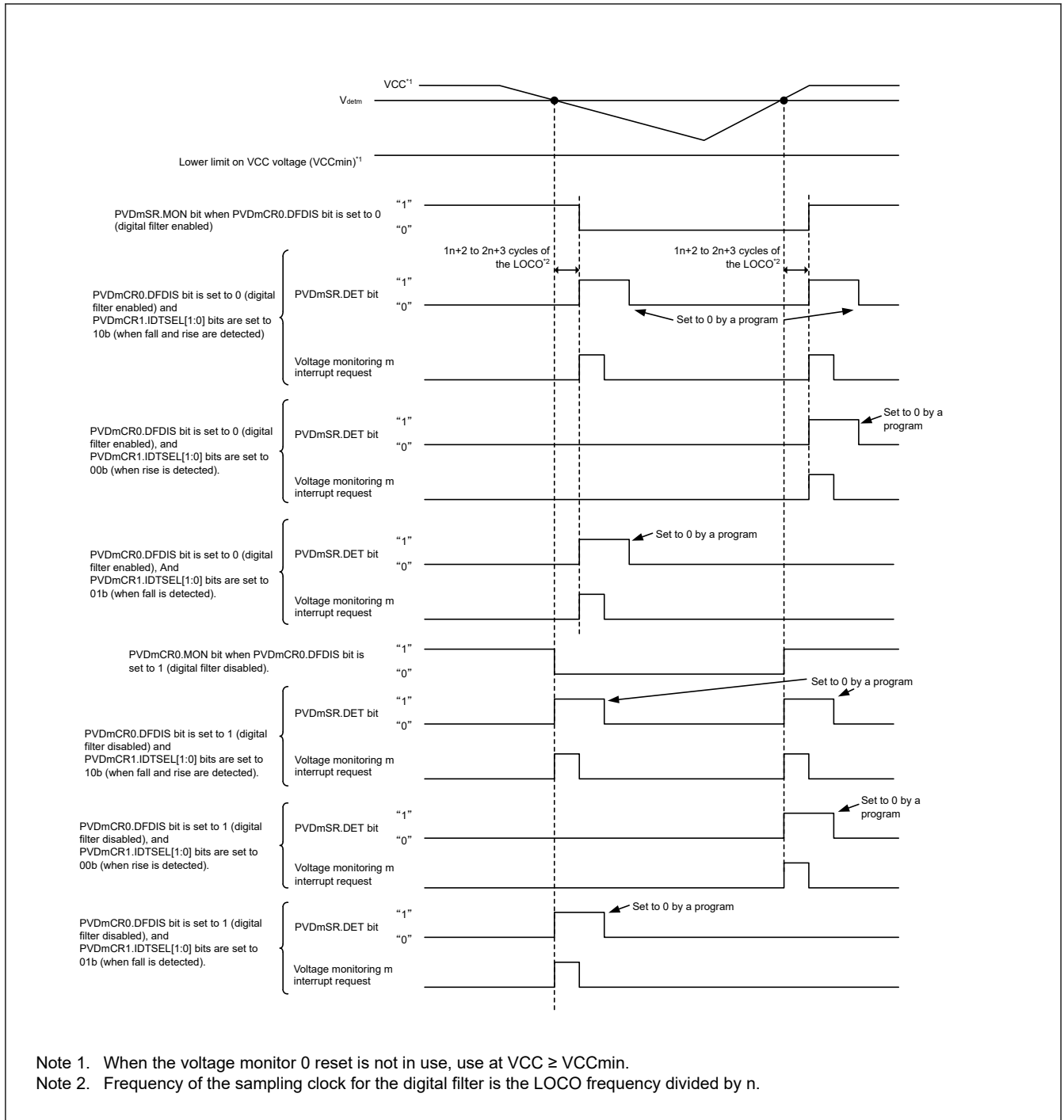


Figure 8.5 Example of voltage monitor m interrupt operation (m = 1, 2)

8.6 Reset from Voltage Monitor n = 4, 5

A reset can be generated in response to the results of comparison by the voltage detection n circuit.

Table 6.6 lists the procedures for setting bits related to the voltage monitor n reset so that voltage monitoring operates. Table 6.7 shows the procedure for setting bits related to the voltage monitor n reset so that voltage monitoring stops. For the operation of the voltage monitor n reset, see [section 6.3.3. Voltage Monitor Reset](#).

Furthermore, if you intend to use the voltage Monitor n circuit in software standby or deep software standby mode, make settings for the voltage Monitor n circuit according to the following procedures

1. Setting in software standby mode

- Disable the digital filter ($PVDnCR0.DFDIS = 1$).

- After $VCC > V_{detn}$ is detected, negate the voltage Monitor n reset signal following a stabilization time.
2. Settings in deep software standby mode
- Disable the digital filter ($PVDnCR0.DFDIS = 1$)
 - During deep software standby mode 3, the voltage Monitor n circuit is stopped. If you intend to use the voltage Monitor n circuit in deep software standby mode, do not transit to deep software standby mode 3.

Table 8.6 Procedures for setting bits related to the voltage monitor n reset so that voltage monitoring operates

Procedure	Step	Voltage Monitor n Reset
Setting the voltage detection n circuit	1	Set $PVDnCMPCR.PVDE = 0$ (disable voltage detection n before writing to $PVDnCMPCR.PVDLVL$).
	2	Select the detection voltage by setting the $PVDnCMPCR.PVDLVL[4:0]$ bits.
	3	Select the type of the reset condition by setting the $PVDnFCR.RHSEL$ bit.
	4	Set $PVDnCMPCR.PVDE = 1$ (enabling the voltage detection n circuit).
	5	Wait for at least t_d (E-A). ^{*1}
Setting the digital filter ^{*2}	6	Select the sampling clock for the digital filter by setting the $PVDnCR0.FSAMP[1:0]$ bits.
	7	Set $PVDnCR0.DFDIS = 0$ (enabling the digital filter).
	8	Wait for at least $2s + 3$ cycles of the LOCO (where $s = 2, 4, 8, 16$, and the sampling clock for the digital filter is the LOCO frequency-divided by s). ^{*3}
Enabling output	9	Set $PVDnCR0.RE = 1$ (enabling the voltage Monitor n reset).
	10	Set $PVDnCR0.CMPE = 1$ (enabling output of the results of comparison by voltage Monitor n).

Note 1. Steps 5 to 10 can be performed during the waiting time of step 4.

Note 2. Steps 5 to 8 are not required if the digital filter is not in use.

Note 3. Steps 9 can be performed during the waiting time of step 8.

Table 8.7 Procedures for setting bits related to the voltage monitor n reset so that voltage monitoring stops

Procedure	Step	Voltage Monitor n Reset
Settings to stop enabling of output	1	Set $PVDnCR0.CMPE = 0$ (disabling output of the results of comparison by voltage Monitor n).
	2	Wait for at least $2s + 3$ cycles of the LOCO (where $s = 2, 4, 8, 16$, and the sampling clock for the digital filter is the LOCO frequency-divided by s). ^{*1}
	3	Set $PVDnCR0.RIE = 0$ (disabling the voltage Monitor n reset).
Stopping the digital filter	4	Set $PVDnCR0.DFDIS = 1$ (disabling the digital filter). ^{*1*2}
Stopping the voltage detection n circuit	5	Set $PVDnCMPCR.PVDE = 0$ (disabling the voltage detection n circuit).

Note 1. Steps 2 and 4 are not required if the digital filter is not in use.

Note 2. To disable the digital filter from its enabled state and then re-enable it, disable it and wait for at least two cycles of the LOCO before re-enabling it.

If the voltage Monitor n reset setting is to be made again after it has been used and stopped once, the following steps in the procedures for stopping and making the setting can be omitted according to the condition.

- Setting or stopping the voltage detection n circuit is not required if the setting for the voltage detection n circuit is not to be changed.
- Setting or stopping the digital filter is not required if the setting for the digital filter is not to be changed.
- Setting the voltage Monitor n interrupt or reset is not required if the setting for the voltage Monitor m interrupt or voltage Monitor n reset is not to be changed.

8.7 Event Link Controller (ELC) Output

The $PVDm$ ($m = 1, 2$) can output the event signals to the Event Link Controller (ELC).

(1) V_{detm} passage detection event

The PVDm outputs the event signal when it is detected that the voltage has passed the V_{detm} voltage while both the voltage detection m circuit and the voltage monitor m circuit comparison result output are enabled.

When enabling the event link output function of the PVDm, be sure to specify settings to enable the PVDm before enabling the PVDm event link function of the ELC. To stop the event link output function of the PVDm, be sure to specify settings to stop the PVDm before disabling the PVDm event link function of the ELC.

8.7.1 Interrupt Handling and Event Linking

Each PVD has bits to separately enable or disable the voltage monitor m interrupts. When an interrupt source is generated and the interrupt is enabled by the interrupt enable bit, the interrupt signal is output to the CPU.

On the contrary, as soon as an interrupt source is generated, the event link signal is output as the event signal to the other module through the ELC regardless of the state of the interrupt enable bit.

It is possible to output voltage monitor m interrupts in Software Standby mode, Deep Software Standby mode 1, and Deep Software Standby mode 2. The event signals for the ELC in Software Standby mode, Deep Software Standby mode 1, and Deep Software Standby mode 2 are output as follows:

- When the event V_{detm} is detected in Software Standby mode, no event signals are generated for the ELC because no clock is presented in Software Standby mode. Because V_{detm} passage detection flag is preserved, when the supply of the clock is resumed after restoring from Software Standby mode, the event signals for the ELC is output according to the state of the V_{detm} passage detection flag.
- If event of passing V_{detm} is detected in Deep Software Standby mode, no event signal is generated for the ELC.